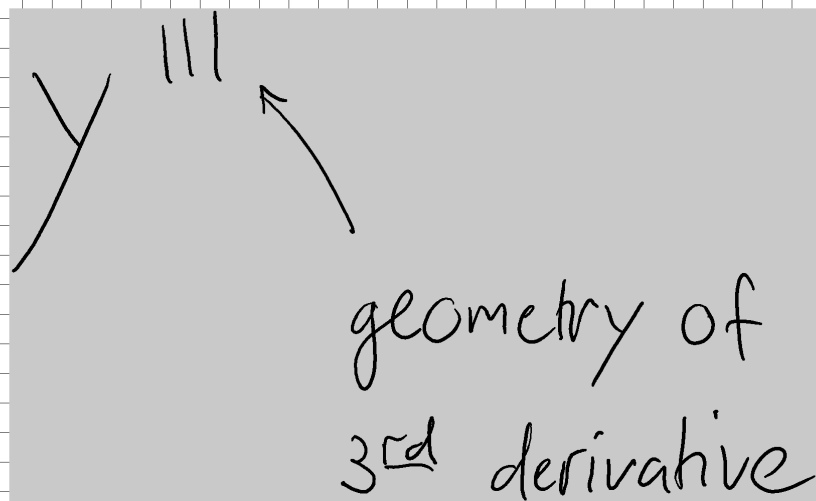




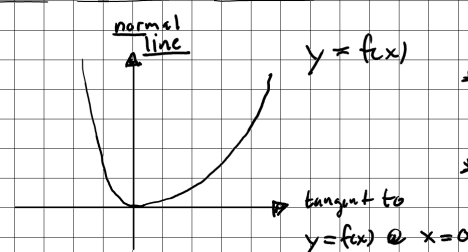
What does the 3rd derivative measure?

note: • slope = y'
how far a curve
is from being a
constant.

• curvature = $\frac{y''}{(1+y'^2)^{3/2}}$
how far a curve is from being a line.

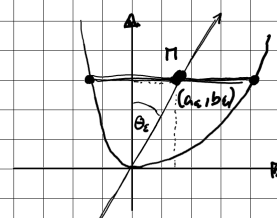
The Aberrancy of Plane Curves

R Gordon, N. Gazette 2005



+ transform so that $f(0) = 0$

* assume $f''(0) \neq 0$



* ϵ = distance from chord to tangent

Define: Aberrancy of $y = f(x)$ at $x=c(=0)$

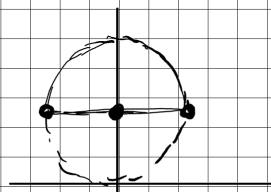
as $\lim_{\epsilon \rightarrow 0^+} \tan(\theta_\epsilon)$

also define: $S_c = \lim_{\epsilon \rightarrow 0^+} \frac{b_\epsilon}{a_\epsilon}$

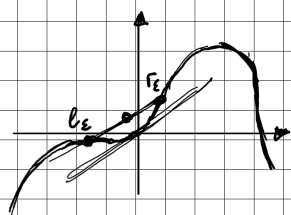
note: $A(c) := \frac{\frac{1}{m} - S_c}{1 - \frac{S_c}{m}}$ with $f'(c) = m$.

Remark: • Aberrancy is invariant under rotation

• Aberrancy of a circle is 0.



a formula if $f(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x$ ($c_2 > 0$)



• equation of tangent line @ $x=0$

$$y = c_1 x$$

• equation of the chord is $y = c_1 x + \epsilon$

define l_ϵ = largest negative intersection of $y = c_1 x + \epsilon$.

r_ϵ = smallest positive " " " "

$$(l_\epsilon, c_1 l_\epsilon + \epsilon), (r_\epsilon, c_1 r_\epsilon + \epsilon)$$

$$(a_\epsilon, b_\epsilon) = \left(\frac{l_\epsilon + r_\epsilon}{2}, \frac{c_1(l_\epsilon + r_\epsilon)}{2} + \epsilon \right) \rightsquigarrow S_c = \lim_{\epsilon \rightarrow 0^+} \frac{b_\epsilon}{a_\epsilon} \\ = \lim_{\epsilon \rightarrow 0^+} \left(c_1 + \frac{2\epsilon}{l_\epsilon + r_\epsilon} \right)$$

observe: l_ϵ, r_ϵ are roots of $\frac{1}{c_n} (f(x) - (c_1 x + \epsilon))$

$$= x^n + \frac{c_{n-1}}{c_n} x^{n-1} + \dots + \frac{c_2}{c_n} x^2 - \frac{\epsilon}{c_n} =: g(x)$$

note:

$$g(x) = (x - l_\epsilon)(x - r_\epsilon)(x^{n-2} + d_{n-3}x^{n-3} + \dots + d_0) \quad (d_i = d_i(c))$$

$$\frac{c_{n-1}}{c_n} = d_{n-3} - l_\epsilon - r_\epsilon \Rightarrow \lim_{\epsilon \rightarrow 0^+} d_{n-3} = \frac{c_{n-1}}{c_n}$$

$$\Rightarrow \lim_{\epsilon \rightarrow 0^+} d_k = \frac{c_{k+2}}{c_n}$$

$$l_\epsilon r_\epsilon d_0 = -\frac{\epsilon}{c_n}, \quad l_\epsilon r_\epsilon d_1 = (l_\epsilon + r_\epsilon) d_0$$

$$\Rightarrow l_\epsilon + r_\epsilon = -\frac{\epsilon d_1}{c_n d_0^2} \Rightarrow \lim_{\epsilon \rightarrow 0^+} \frac{l_\epsilon + r_\epsilon}{\epsilon} = -\lim_{\epsilon \rightarrow 0^+} \frac{d_1}{c_n d_0^2}$$

$$= -\frac{c_3/c_n}{c_n (c_2/c_n)^2}$$

$$\rightsquigarrow S_0 = c_1 - \frac{2c_2^2}{c_3}$$

$$A(0) = \frac{\frac{1}{c_n} - S_0}{1 - \frac{S_0}{c_n}} = c_1 - \frac{(1+c_1^2)c_3}{2c_2^2} = f'(0) - \frac{f'''(0)(1+f'(0)^2)}{3f''(0)^2}$$

$$\therefore \underline{\text{Aberrancy}} = \left| y' - \frac{y'''(1+y'^2)}{(y'')^2} \right|$$

How far a curve is from being a circle.